

Rogério Luis Fontanario

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PROFESSIONAL SUMMARY

Electronics Engineer with over 6 years of hands-on technical experience in hardware development, embedded systems, and software engineering. Specialized in microcontroller programming, PCB layout, prototype assembly, and system testing for hydrogen dosing systems, traffic monitoring, and telecommunications. Demonstrated ability in technical team leadership, product validation, and regulatory homologation.

TECHNICAL SKILLS

- **Programming:** C, C++, Python, Assembly, Flutter
- **Hardware & EDA:** PCB Layout & Schematic Design, Altium, KiCad, EAGLE; Microcontrollers (dsPIC, ESP32, AVR, STM32)
- **Testing & Lab Equipment:** Oscilloscopes, Multimeters, Soldering, System Testing, Product Validation, Troubleshooting, Homologation
- **Languages:** Portuguese (Native), English (Advanced), German (Beginner)

PROFESSIONAL EXPERIENCE

Protium Dynamics Ltda | Maringá, PR, Brazil

Product Engineer | Oct 2022 – Feb 2025

- Led electrical product development team in designing hardware and software for hydrogen generation systems applied to diesel engines.
- Developed microcontroller firmware (C/C++) and cross-platform desktop/mobile monitoring applications (Python, Flutter).
- Executed functional prototype testing, electronic component integration, and authoring of technical documentation.

Velsis Sistemas e Tecnologia Viária SA | Curitiba, PR, Brazil

Product Engineer | Mar 2022 – Oct 2022

- Designed hardware architectures, PCBs, and embedded systems (C/C++) for traffic monitoring and dynamic vehicle weighing systems.
- Executed laboratory and field validation testing for system certifications, homologation, and third-party hardware integration.
- Documented engineering processes for cross-functional teams and contributed feature improvements.

Junior Systems Developer Analyst | May 2021 – Feb 2022

- Engineered embedded systems, executing PCB layouts and low-level microcontroller programming.
- Assisted with field installation and hardware integration of road sensors, traffic radar, and dynamic weighing systems.
- Maintained and updated technical documentation for engineering team reference.

Stein Telecom Ltda | Cascavel, PR, Brazil

R&D Analyst | Jan 2017 – Oct 2019

- Developed hardware solutions, updated legacy PCB designs (migrating from Tango to Altium), and assembled functional prototypes.
- Programmed firmware and supportive mobile applications; contributed to off-grid solar power systems for telecommunications.
- Conducted product testing for homologation and supported assembly line and field teams with documentation.

EDUCATION

- **Bachelor of Engineering in Electronic Engineering** | UTFPR, Toledo, PR, Brazil (3,780 hrs) | March 2011 – August 2017
- **Exchange Program in Electrical Engineering** | Hochschule Schmalkalden, Germany | March 2015 – February 2016
- **Electronics Technician** | CEEP Pedro Boaretto Neto, Cascavel, PR, Brazil (1,893 hrs) | July 2009 – December 2010
- **Industrial Electrotechnical Apprenticeship** | SENAI, Cascavel, PR, Brazil (1,200 hrs) | February 2009 – June 2010

PUBLICATIONS & CERTIFICATIONS

- **Publication:** *Technological Qualification of Hydrogen Micro-Addition in Offshore Diesel Engines: The Case of the H2R System* — 4th Brazilian Hydrogen Congress (2025)
- **Publication:** *Fuzzy Control Applied to Front Anti-Collision Systems and Automotive Electronic Brakeforce Distribution* — Brazilian Fuzzy Systems Congress (III CBSF) / Recent Advances in Systems (2014)
- **Certification:** FuelTech Programmable Fuel Injection Level 1, FuelTech Education (2023)
- **Certification:** Mobile App Developer Bootcamp, IGTI (2020)